INCH-POUND

MIL-PRF-39002/3B
14 May 2004
SUPERSEDING
MIL-PRF-39002/3A
1 February 1999

PERFORMANCE SPECIFICATION

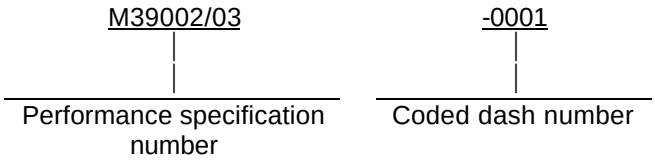
RESISTOR, VARIABLE, WIRE WOUND,
SEMI-PRECISION, STYLE RK11

This specification is approved for use by all Departments
And Agencies of the Departments of Defense.

1. SCOPE

1.1 Scope. This specification covers the associated requirements for a variable, wire-wound, semi-precision resistors. These resistors are capable of full load operation at a maximum ambient temperature of 85°C, and suitable for continuous operation. These resistors have a tolerance of ±5 percent.

1.2 Part or Identifying Number (PIN). Resistors covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded dash number taken from table I. The PIN is in the following form:



2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information are as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC-VAT, Post Office Box 3990, Columbus Ohio 43216-5000, or emailed to resistor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at www.dodssp.daps.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-PRF-39002 - Resistor, Variable, Wire-Wound, Semi-Precision, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202 - Test Method Standard, Electronic and Electrical Component Parts.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or www.dodssp.daps.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-39002.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3 Power rating. The power rating shall be 1.5 watts based on full load operation at an ambient temperature of 85°C.

3.4 Resistance value. Resistance value shall be in accordance with table I.

3.5 Shaft length. Shaft length shall be in accordance with table I.

3.6 Shaft style. Shaft style shall be in accordance with table I.

3.7 Bushing. The bushing shall be locking.

3.8 Nominal resolution. Nominal resolution shall be in accordance with table I.

3.9 Rated working voltage. Rated working voltage shall be in accordance with table I.

3.10 Mechanical travel. The mechanical travel shall be 360°.

3.11 Actual effective electrical travel. The actual effective electrical travel shall be 300° ±3°.

3.13 Electrical output. The electrical output (in terms of percent of applied voltage) is linear with respect to the angular position of the contact arm.

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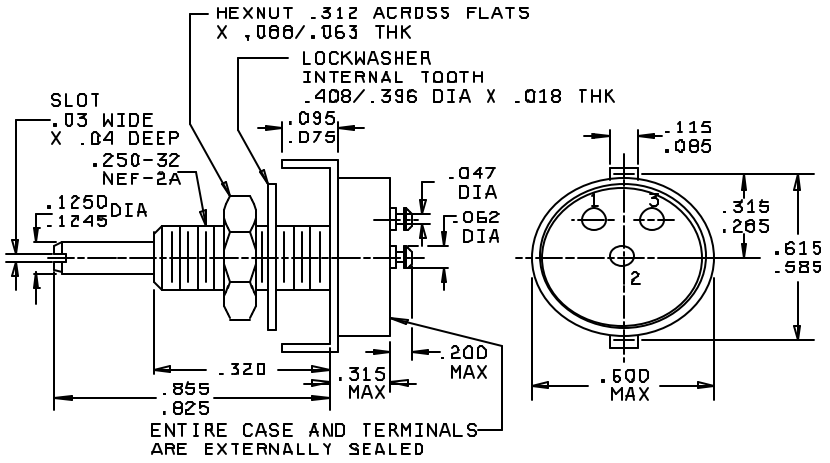
TABLE I. Style RK11 resistor.

Type designation 1/	Shaft length Inches ±.015 shaft style	Nominal total resistance value (ohms)	Nominal resolution 2/ (percent)	Rated ac and dc working voltage (volts)	Terminal	Dash number
RK11SLCS502	.840 inch slotted	5,000 ±5 percent	0.17	122	S	0001

1/ For reference purposes only.
2/ Percent of resolution = 1/N x 100. N = number of turns.

3.12 Hardware. Hardware shall be as specified in figure 1 and shall be corrosion resistant. (Hardware optional between supplier and prime contractor.)



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.018	0.46	.085	2.16	.285	7.24	.585	14.86
.03	0.8	.088	2.24	.312	7.92	.600	15.24
.04	1.0	.095	2.41	.315	8.00	.615	15.62
.047	1.19	.115	2.92	.320	8.13	.825	20.96
.062	1.57	.1245	3.162	.396	10.06	.855	21.72
.063	1.60	.1250	3.175	.408	10.36	4.00	101.60
.075	1.90	.250	6.35				

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Style RK11 resistor.

3.14 Rotational life. The rotational life cycle angle shall be not less than 98 percent of the effective electrical travel angle. Electrical continuity shall be maintained throughout the cycle.

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3.15 Mounting bushing. The mounting bushing shall withstand, without damage, a torque of from 15.5 inch-pounds to 16.5 inch-pounds applied to tighten the mounting nut for five second.

3.16 Shaft to bushing seal (O-ring shaft seal) (see 4.2.1). Shaft to bushing seal shall have no evidence of a continuous stream of bubbles emanating from the shaft to the bushing seal.

3.17 Marking. Due to size limitations, marking of the schematic diagram is optional.

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-PRF-39002.

4.2 Group A inspection. Group A inspection shall include the shaft to bushing seal test.

4.2.1 Shaft to bushing seal. Resistors shall be tested in accordance with following:

a. Mounting shall be in accordance with figure 2. the mounting nut shall be torque to 15.5 inch-pounds to 16.5 inch-pounds.

b. Condition the sample at –62C and +71°C for 15 minutes at each temperature extreme with five minutes maximum between temperature extremes and return to room temperatures.

c. Immerse the test chamber in water as shown in figure 2. Pressurize the sealed test chamber with dry air or nitrogen to 5 psig ±0.5 psig for 60 seconds ±10 seconds.

d. No failures allowed.

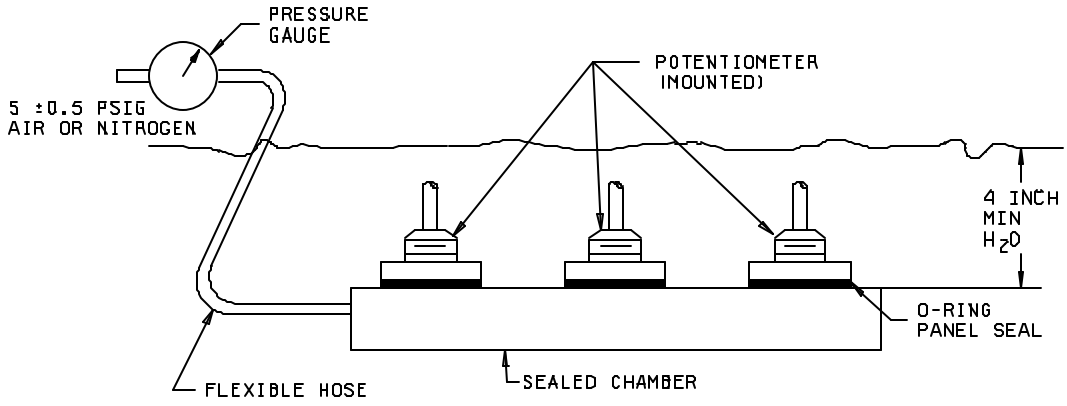

FIGURE 2. Shaft to bushing seal test.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-39002 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, and complete PIN (see 1.2).

b. If not otherwise specified (see 2.1), the versions of the individual documents referenced will be those in effect on the date of release of the solicitation.

c. Packaging requirements.

6.3 Type designation. The type designation for identifying these parts is as follows:

NOTE: This is for information only. For the correct PIN, see 1.2.

RK11	SL	C	S	101
Style (6.3.1)	Shaft and type mounting (6.3.2)	Resistance temperature characteristic (6.3.3)	Terminal (6.3.4)	Resistance (6.3.5)

6.3.1 Style. The style is identified by the two letter symbol “RK” followed by a two digit number; the letters identify semi-precision wire-wound, variable resistors, and the number identified the physical size.

6.3.2 Shaft and type mounting. The style of shaft, type of mounting, and length of shaft is identified by a two letter symbol. The first letter indicates the style of shaft and type of mounting and the second letter indicates the length of shaft (see MIL-PRF-39002).

6.3.3 Resistance temperature characteristic. The resistance temperature characteristic of the completed resistor is identified by a single letter (see MIL-PRF-39002).

6.3.4 Terminals. The terminals are identified by a single letter (see MIL-PRF-39002).

6.3.5 Resistance. The nominal total resistance value expressed in ohms is identified by a three digit number; the first two digits represent significant figures and the last digit specified the number of zeros to follow (see MIL-PRF-39002).

6.4 MIL-PRF-39002/1 (RK09) extended qualification. The manufacturers qualified to style RK09 can extend qualifications to style RK11 by submitting and passing the following additional test:

a. 13 units tested to the strength of mounting bushing test.

b. 13 units tested to the shaft to bushing seal test.

c. No failures are allowed.

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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army – CR
DLA – CC

Review activity:
Army – AR

Preparing activity:
DLA – CC

(Project 5905-2002)

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